



# Bias Is a Math Problem, AI Bias Is a Technical Problem: 10-Year Literature Review of AI/LLM Bias Research Reveals Narrow [Gender-Centric] Conceptions of `Bias`, and Academia-Industry Gap

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## OBJECTIVE AND METHODS

How do researchers in the field of AI/ML fairness operationalize `bias`, and around what axes of identity they most commonly study it?

189 papers across 10 years in \*ACL, NeurIPS, FAccT, and AIES manually screened using questions on defining bias, and proposed implementation

## KEY FINDINGS

### No definition of `bias` [82% papers]

Negative conceptualization of `bias`, conflated with `harm`, without considering how `debiasing` is not always optimal

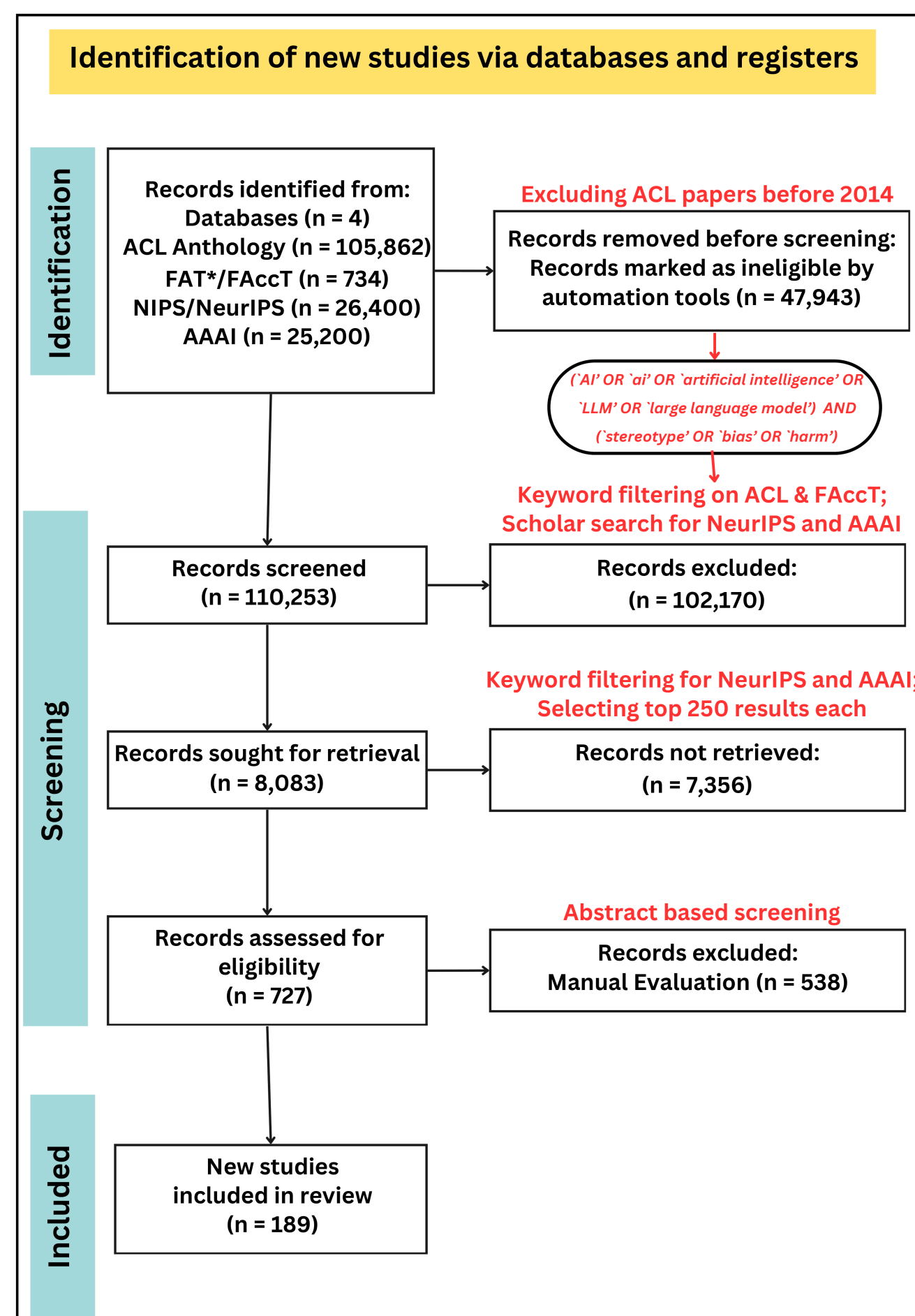
### Overrepresentation of Gender Bias Research [79.9% papers]

30.7% = gender-only, 19.9% = gender-occupation, 18% = intersection of gender & race; 73% = binary gender, but 60% acknowledge limitation

### Absence of Potential Implementation Plans [89.4% papers]

86.2% papers design `novel` debiasing methods, but do not discuss strategies for implementation into real-world AI systems

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| <b>High-level Facts</b> <ul style="list-style-type: none"><li>What aspect(s) of bias is/are mitigated?</li><li>What dataset(s) and model(s) is/are used in experiments?</li><li>How is "success" defined and achieved?</li><li>What methods (qualitative, quantitative, mixed) are used?</li><li>(If human subjects), describe positionality of subjects/annotators.</li></ul>                                                                                                                                                                                                                                                                                              |
| <b>Limitations and Ethical Considerations</b> <ul style="list-style-type: none"><li>What are the stated limitations of this research?</li><li>What potential risks can occur if this work is implemented?</li><li>What subpopulation(s) of users is/are the most likely to experience risks?</li><li>How might potential risks cause harm upon affected users?</li><li>What are some mitigation strategies against the potential risks?</li><li>How will strategies differ across affected subpopulations?</li><li>Has this method been tested for other potential amplification of bias(es)?</li></ul>                                                                     |
| <b>Intended Use Cases</b> <ul style="list-style-type: none"><li>In what real-world systems can this work be implemented?</li><li>What real-world AI systems is this work inappropriate for?</li><li>Is the work applicable for a definition of bias within specific cultures and contexts, or can it be applied in global systems?</li><li>In what languages is this tested, and can it be extended?</li></ul>                                                                                                                                                                                                                                                              |
| <b>Implementation-level Details</b> <ul style="list-style-type: none"><li>What computational resources were required to execute this experiment (e.g., GPU hours, memory, latency tradeoffs, etc.)?</li><li>Have experiments addressed performance in multi-agent settings? How did the implementation of this method affect performance?</li><li>Does the method perform debiasing at training- or inference-time?</li><li>Does the implementation of this method require any retraining baseline model(s) from scratch or modifying any internals?</li><li>Are available experiment details and code clearly documented for independent replication of results?</li></ul> |
| <b>Maintenance and Lifecycle</b> <ul style="list-style-type: none"><li>Are there likely scenarios where this method might break down?</li><li>Are there maintenance tasks that are required for the long-term success of the implementation of this debiasing method?</li><li>Are there components that can expectedly deprecate over time?</li><li>Does the system support human-in-the-loop extensions post-deployment?</li></ul>                                                                                                                                                                                                                                         |
| Contact <researcher name> at <email> for more information!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



## TAKEAWAYS

`Bias` == `bad` promotes purely technical approaches to `debiasing`

Overstudying gender bias can cause harm to other marginalized populations

Lack of implementation plans widens `Academia-Industry` Gap

## RECOMMENDATIONS

Define and Expand focus of `Bias` Coverage

Discuss Implementation Plans via System Fact Sheets

